

Configuration Mapping Specification (CMS)

CMS概述

概述

CMS（通信报文规范）就是国家电网为了摆脱对国外技术的依赖，专门搞出来的“**MMS协议国产替代版**”。它的核心使命是实现电力通信的**自主可控**，同时解决老协议笨重、低效的问题。

- **瘦身提速（PER编码）**：把原来MMS那种臃肿的BER编码换成了紧凑的**PER编码**。
- **砍掉中间商（直接映射）**：以前MMS需要把电力指令层层转译（七层模型），现在CMS**直接映射TCP/IP**。
- **穿上防弹衣（国密加密）**：最大的亮点在于**安全性**。CMS原生支持国密算法加密。

本项目就是制作CMS标准的代码实现。主要的思路是，使用C语言编写per编码部分和基础数据结构部分。后续开发可以基于C/C++，也可以将该模块打包成dll，后续迁移到其他语言平台。项目目录包含下边内容：

- **ccms**
 - per：底层编码，dlt2811协议使用per编码将asn1转换成字节码。
 - data：dlt2811 §7 数据结构部分的实现。
 - svc：dlt2811 §8 报文段部分的实现。
- **jcms-core**
 - core：负责将ccms api映射到java，并且设计基础的 **CmsType** 等基类。
 - info：一些文档性质的枚举和说明。
 - data：dlt2811 §7的java封装。
 - svc：dlt2811 §8的java封装。

- **jcms-utils**

- config: 配置模块。
- scl: scd文件解析模块。
- security: 安全协议。
- transport: 传输层基础构建。

- **jcms-app**

- node: dlt2811 S6传输层封装。
- handler: 各种处理器，比如客户端，服务器，命令行交互界面。
- console: 命令行交互界面。

资料

- 国产自主可控新一代通信标准CMS之总览篇

使用方法

8.1 连接

```
bash
# 非加密连接，默认IP 127.0.0.1、端口 8102。指定访问点会自动
negotiate + associate。
connect --ap C_B5041X/S1;
# 加密连接 (TLS) ，使用 9102 端口
connect --secure --ap C_B5041X/S1;
# 两种连接都可以指定 negotiate 参数，不指定则使用默认值
connect --ap C_B5041X/S1 --apdu 16384 --asdu 65531 --version
1;
# 也可以只进行 TCP 连接，不自动后续操作（没有 negotiate 和
associate)
```

```
connect;  
# 断开连接  
disconnect  
# 退出程序  
exit
```



bash

```
cms> connect --ap C_B5041X/S1;  
Connecting to 127.0.0.1:8102 ...  
Connected, negotiating parameters ...  
Negotiated, associating with C_B5041X/S1 ...  
OK Associated: C_B5041X/S1  
cms> disconnect  
OK Disconnected.  
cms> connect --secure --ap C_B5041X/S1;  
TLS connecting to 127.0.0.1:9102 ...  
Connected, negotiating parameters ...  
Negotiated, associating with C_B5041X/S1 ...  
OK TLS Associated: C_B5041X/S1  
cms> disconnect  
OK Disconnected.  
cms> connect --ap C_B5041X/S1 --apdu 16384 --asdu 65531 --  
version 1;  
Connecting to 127.0.0.1:8102 ...  
Connected, negotiating parameters ...  
Negotiated, associating with C_B5041X/S1 ...  
OK Associated: C_B5041X/S1  
cms> disconnect  
OK Disconnected.  
Negotiated, associating with C_B5041X/S1 ...  
OK Associated: C_B5041X/S1  
cms> disconnect  
OK Disconnected.  
cms> connect  
Connecting to 127.0.0.1:8102 ...  
OK Connected: 127.0.0.1:8102  
cms> disconnect
```

```
Not connected.  
cms> exit  
Bye.
```

8.2.1 确定访问点

```
bash  
# 需要先建立 tcp 连接  
connect;  
# 再指定访问点  
associate --ap C_B5041X/S1;  
# 单独的 associate 可以指定加密（是associate的加密，和connect的加密  
是并行的）  
associate --ap C_B5041X/S1 --secure;
```

```
bash  
cms> connect;  
Connecting to 127.0.0.1:8102 ...  
OK Connected: 127.0.0.1:8102  
cms> associate --ap C_B5041X/S1;  
OK Associated: C_B5041X/S1  
cms> release  
OK Released.  
cms> associate --ap C_B5041X/S1 --secure;  
OK Associated: C_B5041X/S1 (secure)
```

8.2.2 正常释放

```
bash  
# release是正常释放关联，可以后续更换访问点  
release
```

8.2.3 异常释放



bash

```
# 客户端因为某种异常需要断开连接，不需要服务器响应。
# 服务器认为客户端已下线，会关闭 tcp 连接。
# abort <reason>
abort --reason 0; # 默认 reason 就是 0 `others`
```



bash

```
cms> associate --ap C_B5041X/S1 --secure;
    OK   Associated: C_B5041X/S1 (secure)
cms> abort 0
    OK   Abort sent (reason=0)
```

8.3.1 获取逻辑设备



bash

```
# 获取某一个 accesspoint 下边的逻辑设备
server-dir;
# 指定 referenceAfter
server-dir --after LD0;
```



bash

```
cms> connect --ap C_B5041X/S1;
    Connecting to 127.0.0.1:8102 ...
    Connected, negotiating parameters ...
    Negotiated, associating with C_B5041X/S1 ...
    OK   Associated: C_B5041X/S1
cms> server-dir;
    Logical Devices:
        [0] LD0
        [1] MEAS
        [2] CTRL
cms> server-dir --after LD0;
    Logical Devices:
        [0] MEAS
        [1] CTRL
```

8.3.2 获取指定逻辑设备下的所有逻辑节点

```
bash

# LD0 这个设备下的所有逻辑节点
ld-dir --ld LD0;

# LD0 下有很多节点, 获取 LTSM6 之后的节点
ld-dir --ld LD0 --after LTSM6;
```

```
bash

cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> ld-dir --ld LD0;
Logical Nodes:
[0] LLN0
[1] LPHD1
[2] RSYN1
[3] GGI01
...
[75] LTSM5
[76] LTSM6
[77] LTSM7
[78] LTSM8
[79] LTSM9
cms> ld-dir --ld LD0 --after LTSM6;
Logical Nodes:
[0] LTSM7
[1] LTSM8
[2] LTSM9
```

8.3.3 获取逻辑节点目录

```
bash
```

```
# data-object: 数据对象
ln-dir --ln LD0 --acsi data-object;

# data-set: 数据集
ln-dir --ln LD0 --acsi data-set;

# brcb
ln-dir --ln LD0 --acsi brcb;

# urcb
ln-dir --ln LD0 --acsi urcb;

# gocb
ln-dir --ln CTRL --acsi gocb;
```

- data-object
- data-set
- brcb
- urcb
- gocb



bash

```
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> ln-dir --ln LD0 --acsi data-object;
References (data-object):
  [0] Mod
  [1] Mod.stVal
  [2] Mod.q
  [3] Mod.t
  [4] Beh
  [5] Beh.stVal
  [6] Beh.q
  [7] Beh.t
  ...
  [185] Ind1
  [186] Ind1.stVal
```

```

[187] Ind1.q
[188] Ind1.t
[189] Ind2
[190] Ind2.stVal
[191] Ind2.q
[192] Ind2.t
[193] DUTSyn0fs
[194] DUTSyn0fs.stVal
[195] DUTSyn0fs.q
[196] DUTSyn0fs.t
cms> ln-dir --ln LD0 --acsi data-set;
References (data-set):
[0] dsAlarm
[1] dsWarning
[2] dsCommState
[3] dsAin
[4] dsAin1
[5] dsParameter1
[6] dsParameter2
[7] dsParameter5
[8] dsParameter8
cms> ln-dir --ln LD0 --acsi brcb;
References (brcb):
[0] brcbAlarm
[1] brcbWarning
[2] brcbCommState
cms> ln-dir --ln LD0 --acsi urcb;
References (urcb):
[0] urcbAin
[1] urcbAinA
cms> ln-dir --ln CTRL --acsi gocb
References (gocb):
[0] gocb0

```

- lcb
- log


```

cms> connect --ap P_B5041A/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with P_B5041A/S1 ...
OK Associated: P_B5041A/S1
cms> ln-dir --ln LD0 --acsi lcb
References (lcb):
[0] lcblog
cms> ln-dir --ln LD0 --acsi log
References (log):
[0] LD0
cms> ln-dir --ln LD0 --acsi sgecb
References (sgecb):
[0]
cms> ln-dir --ln LD0 --acsi msvcb
References (msvcb):
[0]

```

8.3.4 获取全部数据值

```

# 1. 返回有值的内容, 跳过没设置值的内容
# 2. 已经可以正常支持中文, 使用 unicode-string
# 3. 可以进行 FC 筛选, 默认 XX 不筛选
# 4. 可以进行 referenceAfter 筛选
# all-data --ln <LD|LD/LN> [--fc FC] [--after REF]
all-data --ln LD0 --fc XX --after LicIP17
all-data --ln LD0/LLN0;

```

```

cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1

```

```

cms> all-data --ln LD0/LLN0
Data values (13 items):
  [0] Mod [visible-string] status-only
  [1] NamPlt [visible-string] Name Plate
  [2] LEDRs [visible-string] direct-with-normal-security
  [3] Outage [visible-string] Out of range
  [4] CommTest [visible-string] communication for testing
  [5] CommTstMet [visible-string] communication for
measuring test
  [6] U0AlarmEn [visible-string] U0 Alarm enable
  [7] Set3U0 [visible-string] setting for 3U0
  [8] Meter2En [visible-string] Meter2En
  [9] YKXASSO [visible-string] YKXASSO
  [10] TimeZone [visible-string] TimeZone
  [11] SntpAddr [visible-string] SntpAddr
  [12] YCDeadZone [visible-string] YCDeadZone
cms> all-data --ln LD0/LLN0 --after SntpAddr
Data values (1 items):
  [0] YCDeadZone [visible-string] YCDeadZone
cms> all-data --ln LD0/LLN0 --fc CF
Data values (2 items):
  [0] Mod [visible-string] status-only
  [1] LEDRs [visible-string] direct-with-normal-security
cms> all-data --ln LD0/LLN0 --fc MX
Data values (0 items): (empty)
cms> all-data --ln LD0
Data values (616 items):
  [0] Mod [visible-string] status-only
  [1] NamPlt [visible-string] Name Plate
  [2] LEDRs [visible-string] direct-with-normal-security
  ...

```

8.3.5 获取数据定义



bash

```
# all-def --ln <LD|LD/LN> [--fc FC] [--after REF]
```

```
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
```

```
cms> all-def --ln LD0/LLN0
```

```
Data definitions (16 items):
```

```
[0] Mod [structure] cdc=INC
[1] Beh [structure] cdc=INS
[2] Health [structure] cdc=INS
[3] NamPlt [structure] cdc=LPL
[4] Loc [structure] cdc=SPS
[5] LEDRs [structure] cdc=SPC
[6] Outage [structure] cdc=SPS
[7] CommTest [structure] cdc=SPS
[8] CommTstMet [structure] cdc=SPS
[9] U0AlarmEn [structure] cdc=SPG
[10] Set3U0 [structure] cdc=ASG
[11] Meter2En [structure] cdc=SPG
[12] YKXASS0 [structure] cdc=SPG
[13] TimeZone [structure] cdc=ING
[14] SntpAddr [structure] cdc=ING
[15] YCDeadZone [structure] cdc=ASG
```

```
cms> all-def --ln LD0/LLN0 --after SntpAddr
```

```
Data definitions (1 items):
```

```
[0] YCDeadZone [structure] cdc=ASG
```

```
cms> all-def --ln LD0/LLN0 --fc CF
```

```
Data definitions (6 items):
```

```
[0] Mod [structure] cdc=INC
[1] LEDRs [structure] cdc=SPC
[2] Set3U0 [structure] cdc=ASG
[3] TimeZone [structure] cdc=ING
[4] SntpAddr [structure] cdc=ING
[5] YCDeadZone [structure] cdc=ASG
```

```
cms> all-def --ln LD0/LLN0 --fc MX
```

```
Data definitions (0 items): (empty)
```

```

cms> all-def --ln LD0
Data definitions (256 items):
  [0] Mod  [structure]  cdc=INC
  [1] Beh  [structure]  cdc=INS
  [2] Health [structure] cdc=INS
  [3] NamPlt [structure] cdc=LPL
  [4] Loc  [structure]  cdc=SPS
  [5] LEDRs [structure] cdc=SPC
  [6] Outage [structure] cdc=SPS
  [7] CommTest [structure] cdc=SPS
  [8] CommTstMet [structure] cdc=SPS
  ...
  [253] Beh [structure] cdc=INS
  [254] Health [structure] cdc=INS
  [255] NamPlt [structure] cdc=LPL # 因为我设置了做多256后
边的需要再次请求

```

8.3.6 获取控制块的值

```

bash
all-cb --ln LD0 --acsi brcb;

```

```

bash
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> all-cb --ln LD0 --acsi brcb;
Fetching CB values: target=LD0 type=BRCB
CB values (3 items):
  [0] LLN0.brcbAlarm [BRCB]
  [1] LLN0.brcbWarning [BRCB]
  [2] LLN0.brcbCommState [BRCB]
cms> all-cb --ln LD0 --acsi urcb
Fetching CB values: target=LD0 type=URCB

```

```

CB values (2 items):
  [0] LLN0.urcbAin  [URCB]
  [1] LLN0.urcbAinA [URCB]
cms> all-cb --ln LD0 --acsi lcb
  Fetching CB values: target=LD0 type=LCB
  No CB values found
cms> all-cb --ln LD0 --acsi gocb
  Fetching CB values: target=LD0 type=MSVCB
  No CB values found
cms> all-cb --ln LD0 --acsi msvcb
  Fetching CB values: target=LD0 type=MSVCB
  No CB values found
cms> all-cb --ln LD0 --acsi log # 输入错误
  ERR Invalid acsiClass: log. Valid values: brcb, urcb, lcb,
sgecb, gocb, msvcb
cms> all-cb --ln LD0 --acsi sgecb
  Fetching CB values: target=LD0 type=GOCB
CB values (80 items):
  [0] LLN0.LLN0.SG1  [SGECB]
  [1] LPHD1.LPHD1.SG1 [SGECB]
  [2] RSYN1.RSYN1.SG1 [SGECB]
  [3] GGI01.GGI01.SG1 [SGECB]
  [4] GGI02.GGI02.SG1 [SGECB]
  [5] GGI03.GGI03.SG1 [SGECB]
  ...
  [75] LTSM5.LTSM5.SG1 [SGECB]
  [76] LTSM6.LTSM6.SG1 [SGECB]
  [77] LTSM7.LTSM7.SG1 [SGECB]
  [78] LTSM8.LTSM8.SG1 [SGECB]
  [79] LTSM9.LTSM9.SG1 [SGECB]

```

8.4.1 获取指定数据值



bash

```
# 用法: get-data-values --refs "<ref1> <ref2> ..." [--fc FC]
```

```
get-data-values --refs "LD0/GGIO9.AnIn1 LD0/GGIO9.AnIn2
LD0/GGIO1.HostTPortAlarm";
```

8.4.2 设置指定数据值

```
set-data-values --pairs "<ref1>=<value1> <ref2>=<value2>
...";
```

```
cms> connect --ap C_B5041X/S1;
cms>
cms>
cms> # 数值类型
cms> get-data-values --refs "LD0/LLN0.Mod.stVal
LD0/LLN0.Beh.stVal"
Fetching data values for 2 reference(s)
Data values (2 items):
  [0] LD0/LLN0.Mod.stVal [visible-string] (unavailable)
  [1] LD0/LLN0.Beh.stVal [visible-string] (unavailable)
cms> set-data-values --pairs "LD0/LLN0.Mod.stVal=10
LD0/LLN0.Beh.stVal=20"
Setting 2 data value(s)...
OK Set 2 data value(s) successfully
cms> get-data-values --refs "LD0/LLN0.Mod.stVal
LD0/LLN0.Beh.stVal"
Fetching data values for 2 reference(s)
Data values (2 items):
  [0] LD0/LLN0.Mod.stVal [int32] 10
  [1] LD0/LLN0.Beh.stVal [int32] 20
cms>
cms>
cms> # 浮点数类型
cms> get-data-values --refs "LD0/GGIO2.BusVRtgPri.setMag.f"
Fetching data values for 1 reference(s)
Data values (1 items):
```

```
[0] LD0/GGIO2.BusVRtgPri.setMag.f [visible-string]
(unavailable)
cms> set-data-values --pairs
"LD0/GGIO2.BusVRtgPri.setMag.f=123.45"
Setting 1 data value(s)...
OK Set 1 data value(s) successfully
cms> get-data-values --refs "LD0/GGIO2.BusVRtgPri.setMag.f"
Fetching data values for 1 reference(s)
Data values (1 items):
[0] LD0/GGIO2.BusVRtgPri.setMag.f [float32] 123.45
cms>
cms>
cms> # 布尔类型
cms> get-data-values --refs "LD0/LLN0.CommTstMet.stVal"
Fetching data values for 1 reference(s)
Data values (1 items):
[0] LD0/LLN0.CommTstMet.stVal [visible-string]
(unavailable)
cms> set-data-values --pairs "LD0/LLN0.CommTstMet.stVal=111"
# 错误的类型会被拒绝
Setting 1 data value(s)...
ERR SetDataValues rejected:
cms> set-data-values --pairs "LD0/LLN0.CommTstMet.stVal=true"
Setting 1 data value(s)...
OK Set 1 data value(s) successfully
cms> get-data-values --refs "LD0/LLN0.CommTstMet.stVal"
Fetching data values for 1 reference(s)
Data values (1 items):
[0] LD0/LLN0.CommTstMet.stVal [boolean] true
cms>
cms>
cms> # 字符串类型
cms> get-data-values --refs "LD0/GGIO9.AnIn1
LD0/GGIO1.HostTPortAlarm"
Fetching data values for 2 reference(s)
Data values (2 items):
[0] LD0/GGIO9.AnIn1 [visible-string] (unavailable)
```

```

    [1] LD0/GGI01.HostTPortAlarm [visible-string]
(unavailable)
cms> set-data-values --pairs "LD0/GGI09.AnIn1=光口1发功率修改
LD0/GGI01.HostTPortAlarm=定时信号状态修改"
    Setting 2 data value(s)...
    OK Set 2 data value(s) successfully
cms> get-data-values --refs "LD0/GGI09.AnIn1
LD0/GGI01.HostTPortAlarm"
    Fetching data values for 2 reference(s)
    Data values (2 items):
    [0] LD0/GGI09.AnIn1 [visible-string] 光口1发功率修改
    [1] LD0/GGI01.HostTPortAlarm [visible-string] 定时信号状态
修改

```

8.4.3 获取数据目录

```

bash

# LN 级：列出该 LN 下的所有 DO (不含 fc)
data-dir --ref LD0/LLN0
# DO 级：列出该 DO 下的所有 DA (含 fc)
data-dir --ref LD0/LLN0.Mod
# 分页
data-dir --ref LD0/LLN0 --after Beh

```

```

bash

cms> connect --ap C_B5041X/S1;
cms> data-dir --ref LD0/LLN0
    Data directory (16 items):
    [0] Mod
    [1] NamPlt
    [2] LEDRs
    [3] Outage
    [4] CommTest
    [5] CommTstMet
    [6] U0AlarmEn
    [7] Set3U0

```



```

[8] Meter2En
[9] YKYXASSO
[10] TimeZone
[11] SntpAddr
[12] YCDeadZone
[13] Beh
[14] Health
[15] Loc
cms> data-dir --ref LD0/LLN0.Mod
Data directory (5 items):
[0] [CF]  ctlModel
[1] [DC]  dU
[2] [ST]  stVal
[3] [ST]  q
[4] [ST]  t
cms> data-dir --ref LD0/LLN0 --after Health
Fetching data directory for LD0/LLN0
Data directory (1 items):
[0] Loc
cms> data-dir --ref LD0/LLN0.Mod --after q
Fetching data directory for LD0/LLN0.Mod
Data directory (1 items):
[0] [ST]  t

```

8.4.4 获取数据定义

```

bash
# 用法: get-data-def --refs "<ref1> <ref2>..." [--fc FC]
# DO 级定义 (含 CDC 类型)
get-data-def --refs "LD0/LLN0.Mod LD0/LLN0.Beh"
# DA 级定义
get-data-def --refs "LD0/LLN0.Mod.stVal"
# 带 fc 过滤
get-data-def --refs "LD0/LLN0.Mod" --fc ST

```

```

bash

```

```

cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> get-data-def --refs "LD0/LLN0.Mod LD0/LLN0.Beh"
Fetching data definitions for 2 reference(s)
Data definitions (2 items):
  [0] LD0/LLN0.Mod [structure] cdc=INC
  [1] LD0/LLN0.Beh [structure] cdc=INS
cms> get-data-def --refs "LD0/LLN0.Mod.stVal"
Fetching data definitions for 1 reference(s)
Data definitions (1 items):
  [0] LD0/LLN0.Mod.stVal [boolean]
cms> get-data-def --refs "LD0/LLN0.Mod" --fc ST
Fetching data definitions for 1 reference(s)
Data definitions (1 items):
  [0] LD0/LLN0.Mod [structure] cdc=INC

```

8.5.1 读数据集的值

```

bash
# 用法: get-dataset-values --ds <完整数据集引用> [--after REF]
# 获取数据集所有成员的值
get-dataset-values --ds "LD0/LLN0.dsAlarm"
# 从指定成员之后开始 (分页)
get-dataset-values --ds "LD0/LLN0.dsAlarm" --after
"LD0/GGIO1.Alm1"

```

```

bash
# 具体内容看后边的8.5.5

```

8.5.2 写数据集的值

```

bash

```

```
# 用法: set-dataset-values --ds <ref> --values "<val1>
<val2>..." [--after REF]
set-dataset-values --ds "LD0/LLN0.dsAlarm" --values "true
false true"
```



bash

```
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> set-dataset-values --ds "LD0/LLN0.dsAlarm" --values
"true false true"
cms> get-dataset-values --ds "LD0/LLN0.dsAlarm"
Fetching dataset values for LD0/LLN0.dsAlarm
No dataset values returned
Setting 3 dataset value(s) for LD0/LLN0.dsAlarm
OK Set 3 dataset value(s) successfully
cms> get-dataset-values --ds "LD0/LLN0.dsAlarm"
DataSet values (3 items):
Fetching dataset values for LD0/LLN0.dsAlarm
[0] [boolean] true
[1] [boolean] false
[2] [boolean] true
```

8.5.3 新建数据库



bash

```
# 用法: create-dataset --ds <ref> --members "<ref1>,<fc1>
<ref2>,<fc2>..."
# 创建新数据集
create-dataset --ds "LD0/LLN0.myDs" --members
"LD0/GGIO1.Alm1,ST LD0/GGIO1.Alm2,ST"
# 追加成员到已有数据集
create-dataset --ds "LD0/LLN0.myDs" --members
"LD0/GGIO1.Alm3,ST" --after "LD0/GGIO1.Alm2"
```



bash

案例请看 8.5.4

8.5.4 删除数据库



bash

```
# 用法: delete-dataset --ds <ref>
delete-dataset --ds "LD0/LLN0.myDs"
```

下边这个案例是创建一个新的数据集，然后添加值，查找，删除，确认删除成功



bash

```
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> create-dataset --ds "LD0/LLN0.myDs" --members
"LD0/GGIO1.Alm1,ST LD0/GGIO1.Alm2,ST"
Creating dataset LD0/LLN0.myDs with 2 member(s)
OK Created dataset LD0/LLN0.myDs successfully
cms> get-dataset-dir --ds "LD0/LLN0.myDs"
Fetching dataset directory for LD0/LLN0.myDs
DataSet directory (2 items):
  [0] [ST] LD0/GGIO1.Alm1
  [1] [ST] LD0/GGIO1.Alm2
cms> set-dataset-values --ds "LD0/LLN0.myDs" --values "false
true"
Setting 2 dataset value(s) for LD0/LLN0.myDs
OK Set 2 dataset value(s) successfully
cms> get-dataset-values --ds "LD0/LLN0.myDs"
Fetching dataset values for LD0/LLN0.myDs
DataSet values (2 items):
  [0] [boolean] false
  [1] [boolean] true
cms> delete-dataset --ds "LD0/LLN0.myDs"
```

```
Deleting dataset LD0/LLN0.myDs
OK Deleted dataset LD0/LLN0.myDs successfully
cms> get-dataset-dir --ds "LD0/LLN0.myDs"
Fetching dataset directory for LD0/LLN0.myDs
ERR GetDataSetDirectory rejected: error=1
```

8.5.5 读数据集的成员目录 (key)

```
bash
# 用法: get-dataset-dir --ds <完整数据集引用> [--after REF]
# 列出数据集的所有成员引用 (含 fc)
get-dataset-dir --ds "LD0/LLN0.dsAlarm"
# 从指定成员之后开始 (分页)
get-dataset-dir --ds "LD0/LLN0.dsAlarm" --after
"LD0/GGI01.Alm1"
```

```
bash
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> get-dataset-dir --ds "LD0/LLN0.dsAlarm"
Fetching dataset directory for LD0/LLN0.dsAlarm
DataSet directory (3 items):
  [0] [ST] LD0/GGI01.Alm2
  [1] [ST] LD0/GGI01.Alm1
  [2] [ST] LD0/GGI01.Alm3
cms> get-dataset-values --ds "LD0/LLN0.dsAlarm" # 因为还没有设置值呢, 这三个应该是boolean
Fetching dataset values for LD0/LLN0.dsAlarm
No dataset values returned
cms> get-dataset-dir --ds "LD0/LLN0.dsAlarm" --after
"LD0/GGI01.Alm1"
Fetching dataset directory for LD0/LLN0.dsAlarm
DataSet directory (1 items):
```

```
[0] [ST] LD0/GGIO1.Alm3
cms> get-dataset-values --ds "LD0/LLN0.dsAlarm" --after
"LD0/GGIO1.Alm1"
    Fetching dataset values for LD0/LLN0.dsAlarm # 还没设置值所以
    是空的
    No dataset values returned
```

8.6.1 选择激活定值组

```
bash
# 用法: select-active-sg --ref <sgcbRef> --num <groupNumber>
# 选择定值组 2 为激活组
select-active-sg --ref "LD0/LLN0.SG1" --num 2
```

```
bash
cms> connect --ap C_B5041X/S1; # 连接
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> sgcb-vals --refs "LD0/LLN0.SG1" # 先看
Fetching SGCB values for 1 reference(s)
SGCB values (1 items):
    [0] LD0/LLN0.SG1 numOfSG=4 actSG=1 editSG=1
cms> select-active-sg --ref "LD0/LLN0.SG1" --num 2 # 修改
Selecting active SG: ref=LD0/LLN0.SG1 num=2
OK Active SG set to 2 for LD0/LLN0.SG1
cms> sgcb-vals --refs "LD0/LLN0.SG1" # 再看
Fetching SGCB values for 1 reference(s)
SGCB values (1 items):
    [0] LD0/LLN0.SG1 numOfSG=4 actSG=2 editSG=1
```

8.6.2 选择编辑定值组

```
bash
```

```
# 用法: select-edit-sg --ref <sgcbRef> --num <groupNumber>
select-edit-sg --ref "LD0/LLN0.SG1" --num 2
```



bash

```
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> sgcb-vals --refs "LD0/LLN0.SG1"
Fetching SGCB values for 1 reference(s)
SGCB values (1 items):
[0] LD0/LLN0.SG1 numOfSG=4 actSG=1 editSG=1
cms> select-edit-sg --ref "LD0/LLN0.SG1" --num 2
Selecting edit SG: ref=LD0/LLN0.SG1 num=2
OK Edit SG set to 2 for LD0/LLN0.SG1
cms> sgcb-vals --refs "LD0/LLN0.SG1"
Fetching SGCB values for 1 reference(s)
SGCB values (1 items):
[0] LD0/LLN0.SG1 numOfSG=4 actSG=1 editSG=2
```

8.6.6 读定值组控制块值



bash

```
get-edit-sg --refs "PROT/OCPTOC2.StrVal"
```



bash

具体内容看下边8.6.6

8.6.5 设置定值组值



bash

```
set-edit-sg --refs "PROT/OCPTOC2.StrVal" --values "100" --
type int32
```



bash

具体内容看下边8.6.6

8.6.6 定制组生效 (confirm)



bash

```
confirm-edit-sg --ref "PROT/DeZonePTOC1.SG1"
```



bash

```
cms> connect --ap P_B5041A/S1
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with P_B5041A/S1 ...
OK Associated: P_B5041A/S1
cms> select-edit-sg --ref "PROT/DeZonePTOC1.SG1" --num 1
Selecting edit SG: ref=PROT/DeZonePTOC1.SG1 num=1
OK Edit SG set to 1 for PROT/DeZonePTOC1.SG1
cms> set-edit-sg --refs "PROT/OCPTOC2.StrVal" --values "100"
--type int32
Setting edit SG values (int32) for 1 ref(s)
OK Edit SG values set successfully
cms> get-edit-sg --refs "PROT/OCPTOC2.StrVal"
Fetching edit SG values (SG) for 1 ref(s)
Edit SG values (1 items):
[0] PROT/OCPTOC2.StrVal [visible-string] (unavailable)
cms> confirm-edit-sg --ref "PROT/DeZonePTOC1.SG1"
Confirming edit SG values: ref=PROT/DeZonePTOC1.SG1
OK Edit SG values confirmed for PROT/DeZonePTOC1.SG1
cms> get-edit-sg --refs "PROT/OCPTOC2.StrVal"
Fetching edit SG values (SG) for 1 ref(s)
Edit SG values (1 items):
[0] PROT/OCPTOC2.StrVal [int32] 100
```

8.6.7 定制组信息查看



bash

```
# 用法: sgcb-vals --refs "<ref1> <ref2>..."
sgcb-vals --refs "LD0/LLN0.SG1"
```



bash

```
cms> connect --ap C_B5041X/S1;
Connecting to 127.0.0.1:8102 ...
Connected, negotiating parameters ...
Negotiated, associating with C_B5041X/S1 ...
OK Associated: C_B5041X/S1
cms> sgcb-vals --refs "LD0/LLN0.SG1"
Fetching SGCB values for 1 reference(s)
SGCB values (1 items):
[0] LD0/LLN0.SG1 numOfSG=4 actSG=1 editSG=1
```

8.7

8.15 协商



bash

```
# 使用默认参数（从配置文件读取）
negotiate
# 手动指定参数
negotiate 16384 65531 1;
```



bash

```
cms> negotiate
OK Negotiate completed.
cms> negotiate 16384 65531 1;
OK Negotiate completed.
```

8.16 测试ip



bash

```
# 也可以用来实现心跳机制
```

```
test
```

```
bash
```

```
cms> test
```

```
OK Ping/pong OK
```

```
---
```

```
## ccms
```

```
### 使用方法
```

本模块主要的功能是封装基础数据结构并提供对应的编码解码功能，采用c语言编写。可以将本模块打包成动态库供后续开发使用，也可以基于本模块进一步开发。

```
#### 构建要求
```

- **CMake** (≥ 3.14)
- **C 编译器**: clang 或 gcc
 - Windows 推荐 [LLVM-MinGW (UCRT)](<https://github.com/mstorsjo/llvm-mingw>)
 - macOS 可通过 `'xcode-select --install'` 安装
 - Linux 可通过系统包管理器安装

```
#### 构建产物
```

平台	输入产物	打包后名称
-----	-----	-----
Windows	<code>'libccms.dll'</code>	<code>'ccms.dll'</code>
macOS	<code>'libccms.dylib'</code>	<code>'ccms.dylib'</code>
Linux	<code>'libccms.so'</code>	<code>'ccms.so'</code>

> 去除 `'lib'` 前缀是为了适配 JNA 的 `'Native.load("ccms", ...)'` 加载约定。

```
#### 构建命令
```

```

```bash
Windows (PowerShell)
cd ccms
.\win_ccms.ps1

macOS / Linux
cd ccms
chmod +x ccms.sh
./ccms.sh

```

构建完成后产物输出至 `ccms/dist/` 目录。

## per

- **INTEGER** ( `cms_integer.h` )
  - 没有前导码、没有 Tag、没有自带的 Length；是否对齐，只看它的取值范围 ( `lb..ub` )
  - 对于 2811，小的不需要对齐（比如 `Boolean`、`Quality`），其他的都需要对齐（因为 `range > 255`）
  - 不需要对齐的举例： `Boolean ::= INTEGER (0..1)`、`SmpMod ::= INTEGER (0..2)`
  - 需要对齐的举例： `Int16U ::= INTEGER (0..65535)`
- **OCTET STRING** ( `cms_string.h` )
  - 永远都是对齐，长度字段视情况而定。
  - `OCTET STRING (SIZE(n))`：固定长度。不需要编码长度，因为长度已在 ASN.1 文件中规定。
  - `OCTET STRING (SIZE(lb..ub))`, `ub < 64K`：变长。长度为字符串实际长度。若长度超过 64K，长度编码会越界，必须改用不受约束长度决定因子。
  - 无约束（只在 Data 中有一个）：需要对齐（实际与带约束的编码一样，只是不检查长度）。

- **BIT STRING** ( `cms_string.h` )
  - **BIT STRING (SIZE(n))** : 固定 BIT STRING,  $n \leq 16 \rightarrow$  不对齐;  $n > 16$  或变长  $\rightarrow$  对齐。
  - **BIT STRING (SIZE(lb..ub))**,  $ub < 64K$  : 需要对齐, 需要编码长度, 长度是 **bit 数**。
  - 无约束 (只在 Data 中有一个) : 需要对齐 (实际与带约束的编码一样, 只是不检查长度)。
- **VisibleString** ( `cms_string.h` )
  - 已知倍数字符串, 1 字符 = 1 字节。
  - 约束不编码; 只有实际长度 (字符数 = 字节数) 可能编码。
  - 固定 **SIZE(n)** :  $n \times 8 < 16 \rightarrow$  不对齐;  $n \times 8 > 16 \rightarrow$  对齐; 无长度字段。
  - 变长 / 无约束: 先编码长度决定因子, 内容对齐 ( **> 16 bit** 时 )。
- **UTF8String** ( `cms_string.h` )
  - 标准实际上并没有定义, 这里按 VisibleString 实现, 长度也为字节数。仅在 Data 中使用。
- **OPTIONAL** ( `cms_sequence.h` )
  - 提供了一个函数: 若字段为可选, 标准要求在前面加一个 bit 表示该字段是否存在。
- **SEQUENCE OF** ( `cms_sequence.h` )
  - 提供了一个函数: 此类数组需要编码元素个数。
- **工具流** ( `cms_stream.h` )

## | data

- 这个模块主要实现了dlt2811协议第七章的数据结构，
  - 7.1基础数值类型（basic）：包括了布尔类型、整数、浮点数、字符串等类型。
  - 7.2扩展数值类型（extended）：包括了两种时间。
  - 7.3公共基础类型（common）：包括了比如双点位置(Dbpos)、品质(Quality)等专门对应dlt2811场景语义信息的结构。
  - 7.4功能约束（fc）：只包含功能约束一个内容。
  - 7.5控制块相关（control）：控制块是某一种报文的整体的一组数据结构，这种数据结构本身有一些配套的信息，比如控制操作的发出者(Originator)、控制操作的检测(Check)、控制操作的附加原因(AddCause)。
  - 7.6控制块（block）：各种控制块本身、触发原因以及对应控制块的选项域。
  - 7.7数据union以及数据类型（data）：Data是一个union，用于表达某一种类型的数据。DataDefinition用于表达类型本身。
- 主要包含两种api
  - 提供给内部使用的api。它接收的参数是流的结构，这样可以避免频繁创建新的流。因为可能有一些组合类型需要多个基础类型组合起来依次编码解码。
  - 提供给外部使用的api。它编码导出的是byte数组，主要提供给外部作为一个整体使用。

## svc

ccms主要进行的是基础数据结构的构建。所以svc（service）模块虽然负责对第八章的数据结构进行编码解码，但是主要负责的是对数据结构的编码解码，而不是对协议报文段本身的实现。